HTTP protocol review

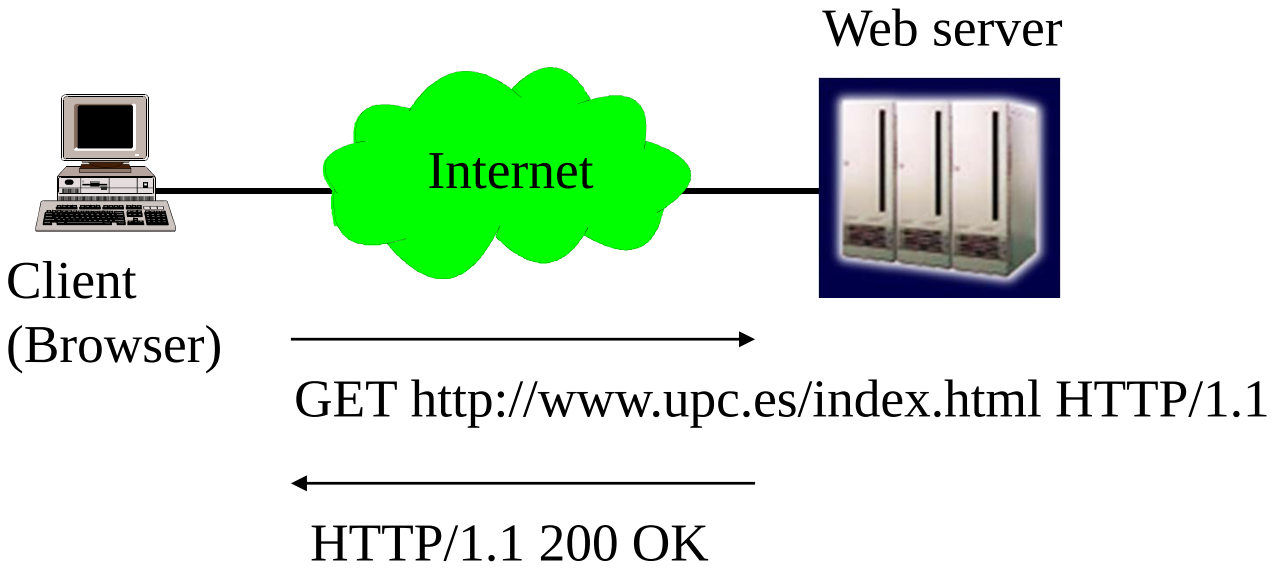
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Communication Protocols

- Transport protocols:
  - TCP/IP (RPC), UDP/IP (MOM)
- Service protocols:
  - FTP (File Transfer Protocol)
  - TELNET (Remote connection)
  - **HTTP** (Hypertext Transfer Protocol)
  - POP3, IMAP
  - SMTP (Simple Mail Transfer Protocol)

The HTTP service protocol

- HTTP/1.1: RFC-2616
  - Request/reply protocol



Protocol phases

- TCP/IP connection set up
- Client request
- Server reply
- ...
- Client request
- Server reply
- Connection tear down

HTTP - request

GET http://localhost/index.html HTTP/1.1

Accept: */*

Accept-Language: en-us

Accept-Encoding: gzip, deflate

User-Agent: Mozilla/4.0 (compatible; MSIE 4.01; Windows NT)

Host: juanluis.upc.es

Proxy-Connection: Keep-Alive

Cookie: ScriptLanguagePreference=VBScript; ComponentPreference=VB5;
SITESERVER=ID=a9398bbd1fcb86bbbe193585d6ef5478

HTTP - reply

HTTP/1.1 200 OK
Server: Apache/1.3.9 (Win32) tomcat/1.0
Last-Modified: Wed, 20 May 2010 06:59:42 GMT
ETag: "0-689-35627f5e"
Content-Length: 1673
Content-Type: text/html

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 3.2 Final//EN">
<HTML>
<HEAD>
<TITLE>Test Page for Apache Installation on Web Site</TITLE>
</HEAD>

Request methods

- **OPTIONS**: asks for the communication options.
- **GET**: obtains the requested information for the requested-URI.
- **HEAD**: the same as on GET but without contents reply.
- **POST**: the same as on GET but request data is attached on the message’s body.
- **PUT**: save on the server the message’s body to the requested-URI.
- **DELETE**: delete the file of the requested-URI on the server
- **TRACE**: the content of the reply is the client request

HTTP session management

- HTTP is a stateless protocol:
  - There is no conversational behavior
  - There are no persistent data on the HTTP packets
- To set up a statefull protocol we must use:
  - Cookies
  - Query String

Cookies

- What’s a cookie?
  - Message-header on HTTP response sent from the server to the browser
  - It contains specific key-values from a particular web application
  - It can be rejected by the browser
  - It may be a security risk identifying a user by her associated cookies
  - A single reply may contain several cookies

Query Strings

http://HostName:PortNumber/theServlet/path?p1=val1

- The query string starts with “?”
- It uses pairs of the form name = value for different parameters
- Each pair name = value is delimited by an “&”
- It uses “%20” to include white spaces
- HTML forms automatically generate URL query strings including all data on clicking the SUBMIT button
- Any pair name=value may be used as a session identifier

Exercises

- Open the http_exercise with NetBeans.
- Try to understand the HTTP_Client emulating an Internet browser, sending HTTP requests to web sites and printing the text with the received reply.
- Try to understand the HTTP_Server emulating a web server, printing the text with the HTTP request and sending a simple HTML reply.
- Try to modify the code of the HTTP_Client to send different HTTP requests.

Content presentation

- HTML (Hypertext Markup Language)
  - Hypertext documents
  - Includes commands for simple formatting and linking to other documents.
  - Provides internal or external links to other web servers
  - It is derived from the SGML (Standard Generalized Markup Language)

HTML

- An HTML page is a text document made up of:
  - Text to be visualized
  - Formatting commands
  - Commands to insert special objects:
    - Images
    - Audio
    - Video
    - Scripts

Basic Commands (i)

- Headers with different letter sizes: `<hx>`, x:1..6
- Bold `<b>`, Italic `<i>`, underscore `<u>`
- New line: `<br>`
- Paragraph `<p>` … `</p>`
- Image insertion:

```
<img src=“image.gif” alt =“alternative
 text“>
```

- Other document linking:

```
<a href="viewcart.do"> View cart </a>
```

- Modify text font:

```
<font size="2" face="Verdana">
```

Basic commands (ii)

- Tables:

```
<table>
<tr>
  <td> … </td>
  <td> … </td>
  <td> … </td>
</tr>
</table>
```

- Forms:

```
<form action="neworder.do" method="post">
   <input type="hidden“
          name="productid“
          value=“1">
   <input type="hidden"
          name="categoryid"
          value=“12">
   <input type="submit"
          name="submit"
          value="add to cart">
</form>
```